

Hepatotoxicity during 6-thioguanine treatment: protocol for a systematic review

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INTRODUCTION

Rationale

Hepatotoxicity was recognised as a complication of 6-thioguanine (6TG) in 1976¹. Since then, 6-TG associated hepatotoxicity in the form of acute sinusoidal obstruction syndrome (SOS), also known as veno-occlusive disease (VOD), has notably been reported in childhood acute lymphoblastic leukaemia (ALL)²⁻⁴, whereas chronic nodular regenerative hyperplasia (NRH) has been reported in adults with inflammatory bowel disease (IBD)⁵ as well as in childhood ALL⁶. However, SOS and NRH should be considered part of a spectrum of microvascular disorders caused by endothelial injury (*Figure 1*)⁷⁻⁹. Nevertheless, the cellular mechanisms responsible for the sinusoidal damage, being pivotal to their development, remain to be established. 6TG-related SOS and NRH have been hypothesised to be dose-related, since they do not seem to occur with low cumulative doses^{10,11}.

A metaanalysis from 2011 reported a seven-fold increased incidence of SOS in two of three randomised trials of childhood ALL, where patients in the experimental arm received 6TG during maintenance treatment in doses of 40-60 mg/m² instead of 6-mercaptopurine (6MP)¹². The meta-analysis highlighted SOS as a dose-related toxicity, as the incidence of SOS in the Childrens Cancer Group-1952 trial was relatively decreased by 20%, when 6TG dose was reduced from 60 to 50 mg/m²³. Furthermore, 6TG-induced SOS was recognised as multifactorial, since the Cooperative ALL trial, in contrast to the two other randomised trials, did not co-administer vincristine and corticosteroids, and that trial had no SOS cases in the 6TG arm, although a high frequency of unexplained thrombocytopenia¹³.

6TG has frequently been used in the management of IBD patients unresponsive to conventional first-line therapy (azathioprine and 6MP) since 2001¹⁴. A non-systematic review from 2013 reported NRH in 2-53% of IBD patients treated with 6TG between 2003 and 2007, however no NRH cases are reported in the period 2008-2011⁵. This review similarly highlighted a presumed dose dependency, since no NRH cases were reported with 6TG doses <20 mg/day⁵. Currently in IBD patients doses <40mg/day are recommended with dose adjustments based on erythrocyte levels of thioguanine nucleotides (Ery-TGN)¹⁵.

Thiopurines were first synthesised in 1951, and are pro-drugs that cause cell death through incorporation of thioguanine nucleotides (TGN) into DNA and subsequently cell death due to futile

repair attempts by the mismatch repair machinery¹⁶. Although 6TG was synthesised before 6MP, its pharmacological properties (including higher toxicity in murine models¹⁷) along with its more challenging synthesis led researchers to focus on 6MP. When 6MP was approved by the FDA in 1953, it completely changed the prognosis of the hitherto deadly acute ALL¹⁸ and is still a cornerstone of childhood ALL treatment¹⁹.

Overall survival for childhood ALL now exceeds 90%²⁰, however the dismal prognosis following relapse still remains a major challenge, with relapse rates of 15-20% and overall survival rates after relapse of 40-70%²¹. Thus, with no new major antileukaemic drugs since the 1970s being part of contemporary first-line ALL therapy²², it is important to explore possible improvements in the use of well-known drugs such as 6TG. The theoretical pharmacological advantages of 6TG over 6MP include a more direct intracellular activation pathway, higher potency, and shorter duration of exposure necessary for cytotoxicity^{3,23}. In this context, the recent finding that higher levels of thioguanine nucleotides incorporated into leucocyte DNA (DNA-TGN) correlate to a lower relapse risk²⁴, calls for reappraisal of the feasibility of prolonged 6TG treatment for childhood ALL, while avoiding the risk of SOS and NRH.

Objectives

- Primary objective: To assess the incidence of hepatotoxicity in patients treated with 6TG compared to 6MP or standard care.
- Secondary objective: To explore if a safe dose of 6TG can be established.

METHODS

The present protocol has been prepared in accordance with the Cochrane Handbook²⁵ and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) statement²⁶. In accordance with guidelines, the protocol will be uploaded to the International Prospective Register of Systematic Reviews (PROSPERO). The systematic review will be conducted in accordance with the Cochrane Handbook²⁵ and PRISMA statement²⁷. The Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach will be used to evaluate the quality of evidence²⁸, and risk of bias will be assessed using the Cochrane Collaboration risk of bias tool²⁹ for randomised controlled trials and the ROBINS-I³⁰ approach for non-randomised studies.

Eligibility criteria

Studies will be selected according to the criteria outlined below.

Types of studies

- Randomised controlled trials (RCTs)
- Controlled clinical trials (CCTs) including quasi-randomised trials
- Observational studies (e.g. prospective and retrospective cohort studies, case-control, and cross-sectional studies)
- Case series and case reports, these will be reported separately

Types of participants

Patients at any age with any disease treated with 6TG. Patients with known liver disease will be subject to sub analyses, as liver disease may entail an inherently higher risk of hepatotoxicity⁸.

Types of interventions

Treatment with 6TG compared to 6MP or standard care (other non-6TG treatment regimen) (comparator only applicable for intervention studies).

Information sources

Electronic searches

We will systematically search the following databases:

- PubMed/MEDLINE (from 1998 to 2018 – the time limit was set at 1998 as 6TG was revisited in childhood ALL treatment in 1998, when two pilot trials compared 6TG to 6MP treatment^{31,32}, as well as to include the full period in which 6TG has been used in IBD treatment, which started in 2001¹⁴)
- Embase/Elsevier (from 1998 to 2018)
- Scopus/Elsevier (from 1998 to 2018)
- Web of Science Core Collection/Clarivate Analytics (from 1998 to 2018)
- The Cochrane Central Register of Controlled Trials (CENTRAL)/Wiley (The Cochrane Library, issue 2, 2018) (from 1998 to 2018)

The search strategies used for the abovementioned electronic databases will be available in appendices to the review.

Other sources

We will hand search the following resources:

- The ‘Similar records’ option in Embase/Ovid, the ‘Related Articles’ option in PubMed/MEDLINE, and the ‘citation tracking’ feature in Web of Science Core Collection/Clarivate Analytics will be used
- The reference lists of included studies
- Conference proceedings from:
 - the American Society of Hematology (ASH) (from 1998 to 2018)
 - European Hematology Association (EHA) (from 1998 to 2018)
 - American College of Gastroenterology (ACG) (from 1998 to 2018)
 - European Crohn’s and Colitis Organization (ECCO) (from 1998 to 2018)
 - International Union of Basic & Clinical Pharmacology (IUPHAR) (from 1998 to 2018)
 - European Association for Clinical Pharmacology and Therapeutics (EACPT) (from 1998 to 2018)
- ClinicalTrials.gov registry (www.clinicaltrials.gov), the International Standard Randomized Controlled Trial Number (ISRCTN) registry (www.controlled-trials.com), WHO International Clinical Trials Registry Platform (ICTRP) (www.who.int/ictrp/search/en)
- PROSPERO will be searched for on-going systematic reviews

The search will be developed and carried out by LNT and CUR.

Data collection and analysis

Literature search results will be uploaded to Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia, as a tool in the process of study selection, and will additionally be used to remove duplicates. We will aim to avoid double counting by juxtaposing author names, location and setting, and sample sizes. Study authors will be contacted by e-mail to resolve any uncertainties.

Selection of studies

Two review authors (LNT and CUR) will independently screen titles and abstracts yielded by the search against the inclusion criteria. Eligible studies will be evaluated in full text by both authors for

final inclusion. Disagreement will be resolved through discussion between LNT and CUR; unresolved disagreement will be adjudicated by KS. Reasons for excluding studies will be recorded. The review authors will not be blinded to the journal titles, study authors or institutions.

Data extraction and management

Two review authors (LNT and CUR) will independently extract data from included studies in duplicate using a standardised extraction form, which will be approved by all authors. Disagreement will be resolved through discussion between LNT and CUR; unresolved disagreement will be adjudicated by KS.

Data items

We will extract information on authors and year of publication, study characteristics (design, blinding, setting, duration, inclusion and exclusion criteria, source of funding, conflict of interest, ethical approvals, key conclusions of the study authors), patient characteristics (number of patients, age, gender, ethnicity, disease, comorbidity, concomitant therapy), details of the interventions (duration of 6TG, dose of 6TG, cumulative dose of 6TG, maximum dose of 6TG, route of administration, ery-TGN levels), comparator (comparator drug, duration of 6MP or other standard care drug, dose of 6MP or other standard care drug, cumulative dose of 6MP or other standard care drug, maximum dose of 6MP or other drug, route of administration), follow-up, and risk of bias.

Outcomes and prioritisation

Primary outcome

- Incidence of any hepatotoxicity reported by the studies as SOS, VOD, NRH, discordant thrombocytopenia, drug-induced liver injury or non-specified hepatotoxicity.

If an incidence rate is not available, we will calculate it using prevalence and total follow-up time as reported in the individual studies.

Because of the lack of standardised definitions of hepatotoxicity, authors of the included studies may not have used the above mentioned terms. To assess further hepatotoxicity we will:

- Report any pathological findings of liver biopsies
- Use the Ponte di Legno (PdL) toxicity working group consensus criteria for SOS, which entail fulfilment of at least three out of the following five criteria: (i) hyperbilirubinaemia; (ii)

hepatomegaly; (iii) ascites; (iv) weight gain 5% or more; (v) thrombocytopenia (transfusion-resistant and/or unexplained by treatment)³³.

- Define hepatotoxicity as an increase of over two times upper normal limit in alanine transaminase or conjugated bilirubin, or a combined increase in aspartate transaminase, alkaline phosphatase, and total bilirubin provided one of them is above two times upper normal limit³⁴.

Since the duration of 6TG treatment in relation to development of hepatotoxicity as well as the time to hepatotoxicity after 6TG exposure are unknown⁶, no restrictions in regard to timing will be used. There will be no restrictions by type of setting.

Secondary outcomes

1. Diagnostic methods: Number of patients who had a liver biopsy, indication for liver biopsy, other diagnostic methods, study conclusions about diagnostic methods.
2. Dose reduction or truncation of 6TG due to hepatotoxicity: Data on how many patients had the 6TG truncated or the dose reduced; doses before and after dose reduction will be extracted.

Risk of bias of individual studies

The Cochrane Collaboration tool for assessing the risk of bias²⁵ will be applied to RCTs included in the review. Two review authors (LNT and CUR) will independently collect information on: (1) sequence generation, (2) allocation concealment, (3) blinding of participants and personnel, (4) blinding of outcome assessors, (5) incomplete outcome data, (6) selective outcome reporting, and (7) other bias including baseline imbalance, early stopping, bias due to financial interest, and academic bias. A judgement of the possible risk of bias on each of the seven domains will be rated as 'low risk' or 'high risk'. In the case of insufficient reporting the risk of bias will be judged as "unclear", and study authors will be contacted for additional information. Reviewers will not be blinded to studies. Disagreement will be resolved through discussion between LNT and CUR; unresolved disagreement will be adjudicated by KS. A graphic illustration of potential bias will be made using the RevMan 5.3 (Review Manager 5.3) software, if possible.

Non-randomised studies included in the review will be evaluated for potential bias using the ROBINS-I tool provided by the Cochrane Collaboration³⁰. Pre-specified confounders include: age, gender, comorbidity (including liver disease) and concomitant use of hepatotoxic drugs. Two review authors (LNT and CUR) will independently collect information on bias due to: (1) confounding, (2)

selection of participants, (3) classification of events, (4) classification of interventions, (5) deviations from intended interventions, (6) missing data, (7) measurement of outcomes, and (8) selection of the reported result. A judgement of the possible risk of bias on each of the eight domains will be rated as ‘low risk’, ‘moderate risk’, ‘serious risk’, ‘critical risk’ or ‘no information’. In the case of insufficient reporting, study authors will be contacted for additional information. Disagreement will be resolved through discussion between LNT and CUR; unresolved disagreement will be adjudicated by KS. Reviewers will not be blinded to studies. Studies with a low risk of bias will be ascribed more weight in the overall findings of the review compared to studies with a higher risk of bias.

Assessment of heterogeneity

A forest plot will be used to summarise the findings of the included studies and assess heterogeneity between studies, the plot will be sorted by study design. Meta-regression analyses will not be performed.

Data synthesis

Based on preliminary searches, we do not expect the studies on this subject to be sufficiently homogenous in terms of design and comparator to perform a quantitative data synthesis; hence, a systematic narrative synthesis of the characteristics and findings of the included studies will be provided with the information presented in text, tables as well as a graphical presentation of SOS/NRH incidence compared to 6TG dose in the included studies.

Meta-biases

We will address publication bias by searching for and including grey literature. To address selective outcome bias, we will compare outcomes between protocols and published reports. Where protocols are not available, we will compare outcomes reported in the methods and results sections. No analyses will be carried out to assess the impact of meta-biases, as we will not provide a quantitative synthesis.

Confidence in cumulative evidence

The Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach²⁸ will be used to assess the confidence in cumulative evidence. Two review authors (LNT and CUR) will independently evaluate the quality of evidence in each of four domains: (1) study design, (2) study quality, (3) consistency, and (4) directness. The evidence on each outcome will be graded as

‘very low’, ‘low’, ‘moderate’ or ‘high’. Disagreement will be resolved through discussion between LNT and CUR; unresolved disagreement will be adjudicated by KS.

Amendments

In the event of protocol amendments, each amendment will be listed with date, description of the change and rationale behind it.

Protocol amendment 1, 20.03.18: During our literature screening we have encountered numerous protocols using 6TG as a part of intensive chemotherapy regimens together with multiple other chemotherapeutics. Since it will not be feasible to evaluate the influence of 6TG in this setting, we have introduced a new exclusion criteria: “6-thioguanine only given as part of an intensive chemotherapy phase concomitant with multiple other chemotherapeutics (ie. induction, consolidation, intensification, reinduction, block therapy, or conditioning regimens for bone marrow transplantation).” We will include the study if 6TG was given as part of maintenance treatment.

Protocol amendment 2, 20.03.18: Amendment to information sources, electronic searches. Embase/Ovid was searched, since the Elsevier interface was not accessible at our centre.

Protocol amendment 3, 27.04.18: Amendment to eligibility criteria, types of participants. After completing the screening process we concluded that the majority of studies using 6TG for an extended period of time either as monotherapy or as part of maintenance treatment was within ALL and IBD, we chose to limit the review to these two disease populations in order to make the review more focused.

Protocol amendment 4, 27.04.18: Amendment to data collection and analysis, data extraction and management. We included two new authors, Sofie Arup and Magnus Strøh Schmidt, in the study group. Two review authors (SA and MSS) will independently extract data from included studies in duplicate using a standardised extraction form, which will be approved by all authors. Disagreement will be resolved through discussion between SA and MSS; unresolved disagreement will be adjudicated by LNT.

Protocol amendment 5, 27.04.18: Amendment to risk of bias of individual studies. All assessments of bias will be carried out independently by SA and MSS. Disagreement will be resolved through discussion between SA and MSS; unresolved disagreement will be adjudicated by LNT.

Protocol amendment 6, 27.04.18: Amendment to confidence in cumulative evidence. The GRADE assessment will be carried out independently by SA and MSS. Disagreement will be resolved through discussion between SA and MSS; unresolved disagreement will be adjudicated by LNT.

Protocol amendment 7, 22.10.18: Instead of ROBINS-I, we decided to use the Cochrane Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies to evaluate the bias of observational studies because it was better suited for the included observational studies which generally did not have both an intervention and comparator group, but were mostly cohort studies.

Protocol amendment 8, 22.10.18: We planned on using the ‘related articles’ option in PubMed/MEDLINE, the ‘similar records’ option in Embase/Ovid, and the ‘citation tracking’ feature in Web of Science Core Collection/Clarivate Analytics, however we could not identify a feasible way to incorporate these measures.

Contributors

LNT and CUR are the guarantors of the review. LNT and KS conceptualised the review. LNT, RHL, TLF, KS and CUR drafted the protocol. The protocol was read and approved by all authors.

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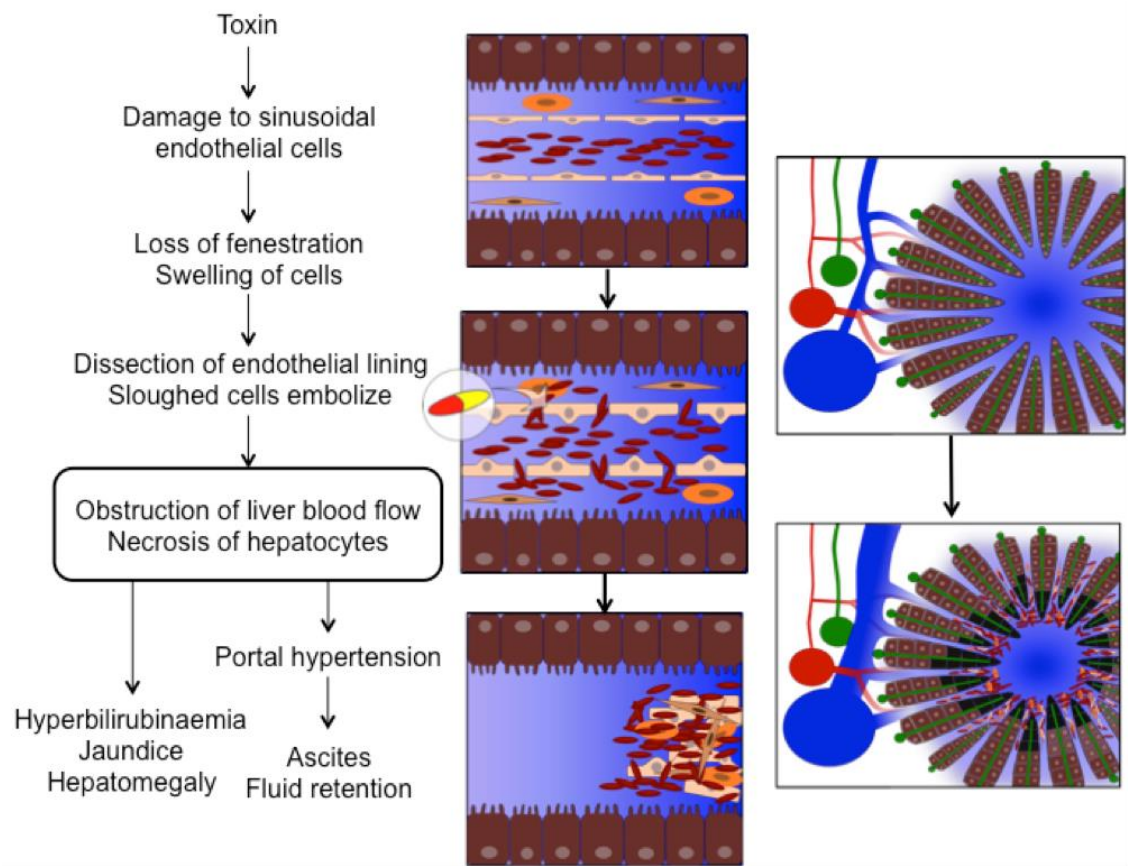
The Danish Childhood Cancer Foundation and the Danish Cancer Society supported this work financially. The funders are not involved in the protocol development, review conduct, data analysis and interpretation, or dissemination of the systematic review.

Competing interests

None declared

Figure 1

Illustration of the pathophysiology of sinusoidal obstruction syndrome



By Linea Natalie Toksvang.

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